MILITARY SPECIFICATION

CRANE: HIGH-SPEED, HIGH-MOBILITY

This specification is approved for use within the U.S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a diesel engine-driven, fully revolving, hydraulic operated, multi-axle, high-speed, high-mobility crane (HSHMC).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

VV-F-800	-	Fuel Oil, Diesel
TT-P-28	-	Paint, Aluminum, Heat Resisting (1200 Deg F.)

MILITARY

MIL-V-173	-	Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronic and Associated Equipment).
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-L-2104	-	Lubricating Oil, Internal Combustion Engine, Tactical Service

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Code LMA-2, Washington, D.C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3810

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MIL-L-2105 - Lubricating Oil, Gear, Multi-Purpose.

MIL-C-3580 - Crane and Crane-Shovels, Truck, Crawler and Wheel-Mounted, Full-Revolving and Their Attachments, Packaging of.

MIL-G-3859 - Grease Gun, Hand-Operated, Lever, Push and Screw Type.

MIL-M-7866 - Molybdenum Disulfide, Technical, Lubrication Grade

MIL-G-10924 - Grease, Automotive and Artillery

MIL-P-14105 - Paint, Heat-Resisting, (For Steel Surfaces)

MIL-S-40626 - Sign Kit, Vehicle Class

MIL-A-46153 - Antifreeze, Etyhlene Glycol, Inhibited, Heavy Duty, Single Package

MIL-L-46167 - Lubricating Oil, Internal Combustion Engine, Arctic.

MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant

MIL-P-52192 - Primer, Coating, Epoxy

MIL-P-53030 - Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free

MIL-C-53039 - Coating, Aliphatic Polyurethane, Single Component Chemical Agent Resistant

MIL-T-62314 - Test Equipment (Simplified) for Internal Combustion Engine (STE/ICE)

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment

MIL-STD-642 - Identification Marking of Combat and Tactical Transport Vehicles

MS24207 - Coupling, Grease Gun, Hydraulic Type Nozzle

MS35000 - Battery, Storage, Lead Acid, Waterproof

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MS51113	-	Switch, Vehicular Lights, 24-volt DC, Waterproof
MS51318	-	Headlight, Blackout, 24-volt, Waterproof
MS51330	-	Stoplight, Taillight, Vehicular - 24 Volt, Blackout Tail, Blackout Stop
MS52125	-	Composite Light, Tail, Stop, Turn and Marker
MS52131	-	Connector, Plug, Electrical Intervehicular Power Cable

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DoT)

Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

US ARMY TANK-AUTOMOTIVE COMMAND

Report No. CR-28-588-003, STE/ICE Design Guide for Vehicle Diagnostic Connector Assemblies

Drawing No. 8383832	-	Tow Bar, Heavy Duty
Drawing No. 8764378	-	Pintle Mount and Assembly
Drawing No. 12258941	-	Connector, Receptacle Electrical (54 Pin)
Drawing No. 12258955	-	Vehicle Test Card

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the

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issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

Safety Standards for Mobile Hydraulic Cranes (ANSI B30.5)

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

POWER CRANE AND SHOVEL ASSOCIATION (PCSA)

Mobile Hydraulic Crane Standards (PCSA No. 4)

(Application for copies should be addressed to the Power Crane and Shovel Association, Marine Plaza, Suite 1700, 111 East Wisconsin Avenue, Milwaukee, WI 53202.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

SAE J371	-	Drain, Fill, and Level Plugs for Earthmoving Machinery
SAE J375A	-	Radius-of-Load and Boom Angle Measuring System
SAE J514	-	Hydraulic Tube Fittings
SAE J516	-	Hydraulic Hose Fittings
SAE J518	-	Hydraulic Flanged Tube, Pipe, and Hose Connections, 4-Bolt Split Flange Type
SAE J534	-	Lubrication Fittings
SAE J551	-	Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (20-1000MHz)
SAE J552	-	External Electromagnetic Radiation Suppressor
SAE J678	-	Speedometers and Tachometers - Automotive
SAE J753	-	Lubrication Chart - Construction and Industrial Machinery
SAE J765	-	Crane Load Stability Test Code
SAE J821	-	Electrical System for Construction and Industrial Machinery

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SAE J931 - Hydraulic Power Circuit Filtration

SAE J994 - Alarm - BackUp - Electric - Performance, Test, and Application

SAE J1024 - Fuel-Fired Heaters - Air Heating - for Construction and Industrial Machinery

SAE J1063 - Cantilevered Boom Crane Structures - Method of Test

SAE J1127 - Battery Cable

SAE J1128 - Low Tension Primary Cable

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

Agent, National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc. Attn: Traffic Department, 1616 P. Street N.W., Washington D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Attn: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D14.3 - Welding Specification for Earthmoving and Construction Equipment.

(Application for copies should be addressed to South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents

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(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2 and 6.3), the contractor shall furnish one crane for first article inspection and approval. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction and manufacturing technique applicable to the remaining items to be furnished under the contract.

3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.3 Design and construction. The crane shall be a high-speed, high-mobility, pneumatic-tired diesel engine driven vehicle, with a fully revolving superstructure and cab, and hydraulically powered telescoping boom. It shall be capable of operating with a concrete bucket, clamshell, grapple and a pile driving and extracting system. The design criteria and practices used, when not specifically specified, shall follow normal commercial crane industry practices for hydraulic cranes. The crane shall be capable of independently and simultaneously lifting and rotating at a speed of not less than 1.5 revolutions per minute. The crane, without load, shall be capable of attaining and maintaining a travel speed of not less than 45 miles per hour (MPH) on paved roads of up to 1 percent grade. The crane, without load, shall be capable of traveling on: highways; unimproved roads; sandy beaches; unprepared and uneven surfaces to include sand, snow and mud. The crane shall be capable of stopping and then accelerating to 1 MPH while on a longitudinal slope of not less than 45 percent in either the forward or reverse direction on paved roads or hard-packed surfaces, and shall be capable of negotiating a 20 percent side slope.

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The crane shall be capable of operating after exposure to chemical decontamination solutions and shall incorporate seals and other synthetic components that are resistant to deteriorating bacteria.

3.3.1 Crane. The crane assembly, designed for carrier mounting, shall consist of a fully revolving superstructure with boom support, hydraulically operated telescoping boom, main winch with wire rope, hook block, cab, controls and necessary accessories.

3.3.1.1 Superstructure. The superstructure shall be mounted on the carrier to insure full working range and designed to withstand the forces caused by lifting and swinging safe loads. It shall not exceed the width of the carrier on which it is mounted. The superstructure shall provide stable retention of the boom in all positions and, with its base, shall rotate on ball thrust or tapered roller bearings of sufficient number, size and capacity to support working loads. Counterweight shall be mounted on the superstructure to provide constant counterbalance to the boom and load.

3.3.1.2 Boom. The boom shall extend to a minimum of 80 feet and when fully retracted, shall not exceed 35 feet. The boom shall be made of alloy steel and shall be a hydraulically operated telescoping type, consisting of a base section and sliding section(s). The boom shall be capable of elevation from the horizontal to a vertical angle of not less than 70°. A boom angle indicator meeting the requirements of SAE J375A shall be furnished. The boom angle indication shall be visible to the operator in his normal seated position. Means shall be provided for the operator to determine the boom length with match points for the load capacity chart. An anti two-block device shall be installed to prevent the hook block from contacting the boom tip.

3.3.1.3 Drum. A cable hoist drum, with wire rope of not less than 3/4 inch diameter, shall be mounted at the base of the boom or on the boom support to facilitate crane operations. The hoist drum shall be actuated by a hydraulic motor controlled from the operator's station. This assembly shall include a motor, a drum and a self-energizing brake mechanism. The winch shall not be free wheeling, and shall be weatherproof for outdoor operation.

3.3.1.4 Hook block. The running block and hook shall have adequate weight to insure payout of cable without load on the hook, with the hook block and boom in any position when reeved for the load which is to be lifted at the given radius. The hook shall be a swivel type with an ultimate breaking strength at least four times the maximum rated capacity of the crane and shall include a safety latch. The hook block shall permit disassembly for visual and nondestructive testing.

3.3.1.5 Rope. Load hoist ropes shall be at least improved plow steel (IPS), independent wire rope center (IWRC), with a safety factor of not less than 3.5. The winch shall be furnished with the manufacturer's standard length rope. The contractor shall provide a signed certification with each crane identifying the crane serial number, wire rope manufacturer, type of wire rope construction, type of wire rope material and the minimum rated breaking strength of the wire rope provided.

3.3.1.6 Sheaves. Sheaves shall have a pitch diameter at least 18 times the rope diameter. The hook block pitch diameter shall be at least 16 times the

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rope diameter. Sheaves shall be adequately guarded to prevent the wire rope from leaving the sheave during operation.

3.3.1.7 Winch. The winch shall have a minimum of two speeds. Low speed shall have a minimum line pull on the last working layer capable of starting a load in mid air equal to 118 percent of the maximum rated capacity divided by the number of parts of line, and obtain a minimum line speed of 110 feet per minute (FPM) at no load. High speed shall have a minimum of 4500 pounds line pull and obtain a line speed of not less than 400 FPM with no load.

3.3.2 Hydraulic system. The hydraulic system shall include, in addition to normal system components, a full flow filter and control valves to insure positive control of boom hoist, boom telescope, rope hoist and outriggers in all operations in event of loss of hydraulic power due to ruptured hoses. Hydraulic line circuit and filtration shall be in accordance with SAE J931. The pump(s) shall be driven by the crane engine and shall have sufficient capacity to operate all hydraulically powered components as specified herein. All rods, which will be exposed during operation, shall have a hard chromium plating. All pressure hydraulic hoses and fitting shall be capable of withstanding a bursting test pressure of four times the working pressure, and a proof pressure of at least two times the operating pressure. When specified (see 6.2), all rods which will be exposed during operation or storage shall be designed to withstand a 20-hour saltwater spray test.

3.3.2.1 Filling, draining and checking provisions. The hydraulic system reservoir, the transmission housing, and all other enclosures which contain lubricant or hydraulic oil shall be equipped with dipsticks, sight gages or checkplugs conforming to SAE J371 to determine the fluid levels. Each enclosure shall be equipped for filling and for draining. Each drain plug or valve shall be located so that removal of the plug or valve will result in complete drainage of the fluid from the enclosure (except torque converter). When the crane is in a level position, drainage of fluids shall be to a container below the crane without draining on or over any part of the crane. Integral tubes, hose or troughs may be used to convey the fluids from the drain outlet. Accessibility to the fill opening, the fluid-level checking device, and the drain plug or valve shall be provided without removal or adjustment of accessories or parts, except for plates equipped with hand-operable, quick disconnect fasteners. The radiator drain shall be such that coolant will not drain onto any part of the crane. Extension hose attached to the draincock is permitted.

3.3.2.2 Ports, fittings and connections. All ports, fittings and connections for sizes larger than 1 inch shall be of the 4 bolt split flange type conforming to SAE J518. For sizes 1 inch or less:

a. Ports shall be the straight thread O-ring type in accordance with SAE J514.

b. Tube fitting shall be 37-degree flared type in accordance with SAE J514, except straight thread O-ring fittings (also in accordance with SAE J514) shall be used when directly connecting to a port.

c. Hose fittings shall be the 37-degree flared swivel type in accordance with SAE J516.

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3.3.2.3 Hydraulic system temperature. Hydraulic tank oil temperature shall not exceed 200°F with an ambient temperature of 120°F while performing repetitive craning operations.

3.3.3 Cabs. The cabs for the carrier and the crane shall be fabricated from sheet steel or fiberglass, having sufficient windows and glazed doors to permit at least 270° vision for the driver or operator in normal position. The cab clearance dimensions for the seated operator shall be based on a 95 percentile U.S. male wearing arctic clothing and safety helmet. All glass shall be safety type, and shall be held in place by rubber or metal channels for easy replacement. Windows shall be arranged for easy opening and removal when required. All cab doors, whether of the sliding or swinging type, shall be adequately restrained from accidentally opening or closing when the crane is traveling or operating. All doors and opening windows shall be provided with means for securely locking in the closed position. Adjustable and comfortable seats shall be provided in each cab, located to provide maximum unobstructed visibility for the operator. Both cabs shall be equipped with air conditioning, heater, defroster, and air or electric operated windshield wiper. An electric or air horn, hand or foot operated, shall be provided in both cabs. The carrier cab shall be provided with West Coast type outside rearview mirrors on each side. A heater with two or more fan speeds and a defroster shall be furnished. The heater shall be capable of reaching and maintaining a uniform cab temperature of 41°F (±9°F) or higher at an ambient temperature of -25°F within 1 hour. The defroster may be either part of a heater-defroster or a separate defroster fan and shall provide defrosting of the front and side windows. If a fan type defroster is provided, it shall be safety guarded, adjustable and provided with an on/off switch. If the heater is fuel fired, it shall conform to SAE J1024. The cabs shall be provided with adequate lighting/illumination to enable the operator to read all gages, charts and control labels under conditions of darkness. The cabs shall have provisions for storage of the operator's M-16 rifle.

3.3.4 Controls. All hoist and crane controls, lockouts and indicators shall be located within easy reach of the operator in the crane cab and shall be readily accessible under all conditions of operation. All controls shall be clearly marked as to the use and function. A positive swing lock for traveling shall be provided. Controls shall be provided in the crane operator's cab to permit remote control of the steering, braking and travel functions of the carrier.

3.3.5 Carrier. Carrier frame shall be of the heavy steel fabrication type with necessary braces and reinforcements to provide rigid support for the revolving superstructure, hoisting drum, rope and hook block under maximum load, when the telescopic boom is extended at minimum through maximum radii at any position within the full working range. The revolving superstructure shall be positioned on the carrier frame to permit maximum efficiency for all purpose crane operations. The crane carrier shall not exceed 102 inches in width. The maximum weight of the completely assembled carrier and crane, without attachments, shall not exceed 75,000 pounds. The number of axles shall be the manufacturer's standard for the capacity crane furnished with all axles to be driven. A differential locking type mechanism shall be provided in the drive line for all driven axles and shall be engaged on demand by the operator using a cab mounted switch. The drive line shall be of the torque proportioning transfer type. The drive line system shall be designed so the operator may

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engage or disengage the front axles while moving without damage to the drive line or other components. Power assist steering for the front wheels shall be provided. The carrier shall have a curb to curb turning radius of no more than 50 feet. The carrier shall be capable of fording 60 inches of saltwater without contaminating the liquid reservoirs or the air induction system. If constant velocity joints are used in the steering axles, they shall have provisions to drain water after a fording operation when required. The carrier with crane shall not exceed 144 inches in height. Two tow lugs, rated at 60 tons, suitable for mating with tow bar conforming to Drawing No. 8383832, shall be furnished and installed on the front and rear of the carrier. A towing pintle conforming to towing pintle mount and assembly (Drawing No. 8764378) shall be furnished on the rear of the carrier.

3.3.5.1 Transmission. The crane carrier shall be equipped with a full power shift or full automatic transmission for the size crane specified. The transmission shall be equipped with a neutral safety start switch. The transmission shall provide adequate gearing for all performance conditions specified herein and operation at any speed up to the maximum speed. The transmission shall be protected from damage when the operator moves or attempts to move the directional control to the opposite direction while the crane is in motion. The transmission shall be compatible with oil conforming to MIL-L-2104, grade 10 and MIL-L-46167. The transmission system shall be furnished with filter(s) having replaceable elements and sized in accordance with the transmission manufacturer's recommendation. The filter shall be replaceable without removal of the pan or other components unless service interval for replacement is the same as for other service requiring removal.

3.3.5.2 Oil sampling valve. An oil sampling valve shall be provided on the engine and transmission. The valve shall be manually operated and shall close automatically after release. It shall be made of material resistant to corrosion such that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located in such a way that samples can be taken easily, without personnel being exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken it shall be a true representation of oil that is flowing when the engine is running. Sampling valves shall be labeled with a plate in accordance with 3.10.2 adjacent to the valve indicating the type of oil sampled (engine or transmission, whichever is appropriate).

3.3.5.3 Brakes. The crane carrier shall be provided with the manufacturer's standard, foot controlled, service brakes applied to all wheels. The service brakes shall be capable of bringing the crane to a complete safe stop on a 45 percent slope, forward or reverse. A parking brake, capable of holding the crane on a longitudinal slope of not less than 20 percent without slippage, shall be provided.

3.3.5.4 Transfer case. If an automatic transmission is provided, the crane shall be equipped with a two-speed transfer case. The transfer case shall be multi-speed, torque proportioning.

3.3.5.5 Tires. Tires shall be of the tubeless type with traction type tread on all axles. High floatation tires shall be provided on the front axles. When specified (see 6.2), all tires shall be high floatation type on

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all axles. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the maximum load. Tire ratings shall conform to Tire and Rim Association (TRA) recommendations for the type and size of tires furnished.

3.3.5.6 Lights and reflectors. The following lights and reflectors shall be furnished and installed and shall be powered by a 24-volt (V) electrical system:

a. A minimum of two sealed beams (mounted on carrier).
b. Taillights.
c. Stoplights.
d. Minimum of eight clearance lights.
e. Turn signal lights.
f. Minimum of three adjustable floodlights mounted on the crane to rotate with the superstructure.
g. Instrument panel lights (both cabs).
h. Emergency 4-way flashing lights.
i. Interior dome lights (both cabs).
j. Reflectors positioned on all corners of the carrier. Yellow reflectors shall be on the left and right sides and front. Red reflectors shall be on the rear of the carrier.

3.3.5.7 Back-up alarm. A back-up alarm shall be provided on the crane and shall comply with SAE J994, Type C. The back-up alarm signal shall be audible above and distinguishable from the surrounding noise level.

3.3.5.8 Engine(s). The crane engine(s) furnished shall be of the diesel type having horsepower, torque and speed characteristics to meet satisfactorily all the crane performance requirements specified herein. The engine(s) shall be capable of meeting the performance requirements using diesel fuel conforming to VV-F-800. The diesel engine(s) shall start within 5 minutes and be ready for full load operation within 15 minutes in any ambient temperature from +125°F to -25°F. When a fluid priming system is required, it shall be of the measured shot type with storage capacity of at least 12 fluid ounces. When specified (see 6.2), the engine(s) shall be capable of starting and operating in temperatures to -40°F with the use of a winterization kit. The engine(s) shall be furnished complete with at least the following accessories:

a. Instrument panels in the cabs complete with a lubricating oil pressure gage, a fuel gage, a cooling liquid-temperature indicator, a voltmeter or ammeter, an hour meter (may be located in engine compartment), and a speedometer with recording odometer and tachometer in the carrier cab.
b. Charging alternator(s) with a capacity rating compatible with the electrical system requirements.
c. Cooling-liquid high temperature visible or audible safety warning device.
d. Lubricating oil low pressure visible or audible safety warning device.
e. Fuel gage.
f. An engine housing designed to facilitate maintenance of the engine.
g. Transmission fluid temperature gage.
h. Air compressor and receiver tank with sufficient capacity to inflate the carrier tires.

3.3.5.9 Fuel system. The fuel system shall be located so that no part of the system extends beyond the widest part of the crane and that no part of the

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fuel tank is forward of the front axle. The fuel tank joints shall be closed by welding, brazing, silver soldering or equivalent technique. Joint closing by crimping or lead-based or soft soldering shall not be used. The capacity of the fuel tank shall be sufficient for not less than 200 miles of highway travel at maximum attainable speeds up to 45 MPH. The fuel tank shall be legibly marked "DIESEL FUEL ONLY" in block letters ½ inch in height. The fill pipe opening shall not be less than 4 inches in inside diameter and shall be located outside of the operator's compartment so that no fuel spilled during fueling will contact any part of the exhaust or electrical systems. The tank filling components shall include a strainer. The fuel tank shall be provided with shut-off valves for all lines to and from the tank. A fuel tank drain plug located at the lowest point shall be furnished. Fuel lines and fittings shall be located and routed in protected locations not subject to chafing or damage. The mounting of the fuel tank shall prevent shifting, damage, and fatigue as a result of travel and transport. The fuel tank and the fuel filter housing shall be equipped with means for draining water and sediment from the fuel system. If the fuel filters are not of the type that separate out water, a water separator shall be provided. Fuel lines shall be sized to provide continuous operation at full throttle and maximum loads.

3.3.5.10 Engine air induction system. The air induction system shall include a pre-cleaner or dust scavenger, a heavy-duty dry type air cleaner and a locking type air cleaner restriction indicator. The air cleaner restriction indicator shall be mounted in a location clearly visible to the operator without requiring removal of panels, covers or lids.

3.3.5.11 Tachometer drive. The engine shall be furnished with a heavy duty key drive tachometer take-off in accordance with SAE J678. The drive shall be 5/32-inch size and shall have an outer threaded male connection with 7/8-18NS-2A threads. The take-off shall be furnished with a treated captive chained cap.

3.3.5.12 Outriggers. Not less than four hydraulically powered outriggers capable of being extended not less than 9 inches below ground level, shall be furnished. The outriggers shall have sufficient strength to accept full stability, under all rated loading conditions, including the ability to raise the crane off the ground from travel position, when necessary, to compensate for ground surfaces having irregular contours. Vertical outrigger cylinders shall have check valves as a part of, or attached to, the cylinder. Horizontally extended hydraulic cylinders shall be enclosed within outrigger boxes. Outriggers in the stowed position, ready for travel, shall not exceed maximum allowable overall width. Devices to accurately determine when the crane is level shall be installed in a position convenient to the outrigger controls. All hydraulic hoses shall be enclosed or routed to preclude external damage.

3.3.6 Electrical system. The crane shall be equipped with a 24-V negative ground, electrical system in accordance with SAE J821, J1127 and J1128.

3.3.6.1 Batteries. The batteries shall be size 6 NT in accordance with MS35000, and shall be readily accessible for servicing. The battery and cable terminals shall be identified with the respective polarity by raised or recessed markings and corrosion resistant nuts and bolts shall be provided.

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3.3.6.2 Battery enclosure. A weather protected box, or compartment-type battery enclosure, shall be provided. The enclosure shall include a drain for water or electrolyte, and a vent to prevent gas accumulation, and shall provide accessibility for servicing without removal of the battery or other components. The enclosure shall be equipped with battery holddown components to prevent any damage to the batteries and shall be internally treated to prevent corrosion.

3.3.6.3 Intervehicular connector. The crane shall be equipped with a 24-V slave receptacle conforming to MS52131. The receptacle shall permit charging the crane batteries and slave starting of the engine from an external source and charging and slave starting other equipment with the crane. The receptacle shall be installed on the exterior of the crane near the battery enclosure and shall be accessible to personnel standing on the ground.

3.3.6.4 Blackout lighting. One blackout headlight conforming to MS51318 shall be mounted as close to the extreme left of the vehicle as practical and positioned to provide illumination with minimum obstruction. Two blackout stoplight-taillights conforming to MS51330 shall be mounted adjacent to the taillights in recesses, or provided with guards. The blackout lights shall be controlled by an interior switch conforming to MS51113 in the carrier cab. All other exterior lights, backup alarm and the horn, if electric, shall be made inoperable automatically when the blackout lights are switched on. A separate wiring harness may be provided for the blackout lights. In lieu of separate service and blackout taillight assemblies, composite light assemblies, conforming to MS52125 may be provided. All interior lighting, including instrument panel lights (exclusive of warning indicator lights) and gages which are illuminated in the blackout mode, shall not emit energy outside of the 380 to 700 nanometer wavelength range. All warning indicators shall be light emitting diodes (LED).

3.3.6.5 Diagnostic connector assembly (DCA). The crane shall incorporate an easily accessible DCA in the operator's cab for interface with the Government's Simplified Test Equipment/Internal Combustion Engine (STE/ICE) in accordance with Report No. CR-82-588-003 and MIL-T-62314, Appendix B. The diagnostic connector shall be in accordance with Drawing No. 12258941. As a minimum, the DCA shall be capable of monitoring the functions defined in Table II of Report No. CR-82-588-003 and alternator/generator output voltage, alternator/generator field voltage, alternator/generator negative cable voltage drop, fuel solenoid voltage, starter voltage, starter current first peak and average, starter circuit resistance, battery internal resistance change, fuel supply pressure and fuel filter restriction (Delta P). When specified (see 6.2), additional test parameters to be monitored in either the DCA or transducer kit (TK) mode shall be specified. Test points identified to monitor functions in the TK mode shall be accessible without the removal of major assemblies and installed attachments. Fittings/adapters required for the interface with the STE/ICE test set when monitoring in the TK mode shall be permanently installed on the crane. Adapters in the STE/ICE test set shall readily mate with test point connections in the crane. A separate wiring harness shall be provided for the DCA and shall include all wiring and necessary hardware to perform required capabilities. The contractor shall also provide STE/ICE vehicle test cards in the format identified in Drawing No. 12258955 addressing both DCA and TK measurements.

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3.3.7 Winterization. A coolant heater, an engine oil heater, and a battery heater shall be provided when specified (see 6.2). Heaters shall operate on 115-V alternating current and shall be wired through a junction block to a single three-pronged (male), weatherproof slave receptacle cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. Connecting cable shall include a matching female connector at the vehicle end and standard weatherproof three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperatures (-40°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Heaters shall be furnished as follows:

a. Coolant heater, 115-V, 1500-watt (W) minimum rating, shall be installed along with a 115-V coolant circulating pump in such manner that heater coolant is continually circulated through the engine block, even when the engine's thermostat is closed. Normally, the pump and heater will take coolant from a port in the lower region of the block or from the lower radiator hose and return it to the head or upper region of the block or from the lower radiator hose and return it to the head or upper region of the block through the cab heater inlet (engine or cab heater) hose. A check valve in the coolant heater circuit shall prevent reverse flow when the engine is running. An engine thermostat with an operating range of 170° to 195°F shall be installed.

b. Immersion type engine oil heater, 115-V, 300-W minimum rating, with a 170° to 195°F thermostat, shall be installed in the oil pan through any convenient opening.

c. Battery heater, 115-V, 200-W minimum rating controlled with an integral 60° to 80°F thermostat shall be installed either under or around the battery in either the manufacturer's standard battery box or one constructed specifically for this purpose. The battery heater shall be suitably protected from the corrosive liquids and fumes normally associated with vehicle batteries.

3.3.8 Tools. All nonstandard or special tools required for operation and maintenance of the crane, which are normally furnished to commercial customers, shall be furnished. In addition, an air hose of sufficient length to inflate the carrier tires using the engine-driven air compressor shall be provided.

3.3.8.1 Toolbox. A permanently mounted, lockable closed compartment or toolbox shall be provided to carry tools and accessories. It shall be of sufficient size to carry all tools provided with the crane. The toolbox shall be made of sheet steel with a minimum thickness of 0.0747 inch (U.S. Standard Gage No. 14). The toolbox shall have a hinged lid with a padlock closing device that will keep the lid closed when the toolbox is subjected to vibration. The toolbox shall be in a protected and accessible location.

3.3.9 Lubrication. Adequate and accessible means of lubricating all surfaces requiring lubrication shall be provided. A lubrication chart identical to the chart provided in the operator and maintenance manuals and in accordance with SAE J753 shall be furnished. The chart shall identify

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applicable military lubricants and appropriate ambient temperature ranges, and shall be mounted on a plate conforming to 3.9.2.

3.3.9.1 Lubricants. All parts and components shall be designed so that they are properly lubricated with specified military lubricants. The engine, transmission and torque converter shall operate on oil conforming to MIL-L-2104, except that for low temperature operation, oil conforming to MIL-L-46167 may be used. Initial lubrication and all subsequent lubrication shall be with military lubricants conforming to MIL-L-2104, MIL-L-2105, MIL-L-46167, MIL-M-7866 and MIL-G-10924. All components shall be filled to the operating level with appropriate operational grades of military lubricants designated for use in the temperature range of the destination of the delivered crane. The crane shall be tagged in a conspicuous place to indicate the temperature ranges and grade of lubrication used.

3.3.9.2 Lubrication fittings. Fittings in accordance with SAE J534 shall be located in protected positions, and shall be accessible to, and compatible with, a grease gun conforming to MIL-G-3859 equipped with a hydraulic coupler conforming to MS24207 and a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts, except that engine-housing side panels and plates equipped with hand-operable, quick disconnect fasteners may be removed.

3.3.9.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure-lubricating equipment will damage grease seals or other parts.

3.3.10 Attachments. The following attachments shall be furnished with the crane, when specified (see 6.2). Changing from the crane configuration to the configuration required for use of any of the attachments shall be accomplished in less than 2 hours.

3.3.10.1 Clamshell. The clamshell shall be a hydraulically operated, general purpose bucket, 1.0 cubic yard capacity, with teeth, controlled from the operator's station. A combination hose reel and tagline reel or a single reel which utilizes the hydraulic operating hose for the tagline function shall be attached to control and actuate the bucket.

3.3.10.2 Grapple. The grapple shall be a hydraulically operated five tyne 1.0 cubic yard grapple with heavy duty cylinders. The cylinders shall provide positive action on all five tynes simultaneously. A combination hose reel and tagline reel or a single reel which utilizes the hydraulic operating hose for the tagline function shall be attached to control and actuate the grapple.

3.3.10.3 Pile driving system. The pile driving system shall be capable of being mounted and controlled by the crane. The pile driving system shall be a complete system for driving and extracting piling using both impact and vibratory means. The system shall consist of one or more hammers or tools capable of driving and extracting various types of piling in unconsolidated and semi-consolidated geological formations. The pile driving system must be operated at a radius of at least 25 feet with the crane on outriggers. The pile driving system will conform to the specification specified in 6.2.

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3.3.10.4 Concrete bucket. The concrete bucket shall have a capacity of 1.0 cubic yard, with a mechanical material release lever that can be operated by one man on the ground.

3.3.11 Vehicle weight classification sign kit. When specified (see 6.2), the contractor shall apply the Government furnished vehicle weight classification numbers to each vehicle utilizing a vehicle weight classification number kit conforming to MIL-S-40626. The kit shall be located on the front of the vehicle in a location approved by the contracting officer. The contracting officer shall assign the classification numbers to be displayed on the vehicle. Instructions for changing the number based on the vehicle configuration shall be provided in the operator's manual.

3.3.12 Fungus and moisture resistance. All electrical components and connections shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition as specified in 3.3.12.1 with 7 percent salicylanilide or 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish, except that:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

Where used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.3.12.1 Composition. Composition I shall be used unless prohibited by local air pollution regulations governing the application of the varnish. When Composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District.

3.3.13 Lifting and tiedown attachments. The crane shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to Type II or Type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the crane. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tiedown of the crane on the carrier when shipped.

3.3.14 Lifting capabilities. The crane shall be capable of lifting 50,000 pounds with the load center, over the side and rear, being a minimum of 5 feet from the fully extended outriggers. The crane shall be compatible with standard commercial spreader bars for use with 20 foot ISO containers.

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3.3.15 Backward stability. The backward stability shall be determined with lubricating and cooling systems full, with fuel tank(s) at least half full, with the crane resting on firm, level and uniform supporting surface, without outriggers touching the surface, with the boom fully retracted and at maximum boom elevation and with the hook block resting on the ground. When in this condition, the minimum weight on the tires under the boom shall not be less than 15 percent of the total weight of the crane when the boom is positioned either parallel or perpendicular to the longitudinal axis of the carrier.

3.3.16 Power load lowering. Power load lowering by hydraulic control shall be provided for the main hoist to facilitate controlled load lowering or lifting at any boom length or elevation. The power load lowering means shall provide positive control of a single line load equal to the maximum rated crane capacity divided by the recommended number of parts of hoist line reeving or maximum permissible line pull, whichever is greater.

3.3.17 Hoist and rotation. The available line pull and speed shall be as rated in accordance with PCSA Standard No. 4 for the operational requirements of the crane hoisting drum. The power transmission system used to rotate the superstructure shall be a hydraulic mechanical arrangement with a continuous swing rotation angle of not less than 360°. Swing speed within this range shall be smooth and constant, and adequate control shall be provided in the hydraulic circuit to insure smooth starting and stopping. The crane shall simultaneously lift and rotate throughout the rated working area.

3.3.18 Rated load capacities chart. A permanent chart indicating all the rated load capacities on each boom reach, and radii with outriggers extended, shall be posted in the crane cab near the operator and also on the outside of the crane to be legible from ground level. The rated load capacity chart shall reflect the manufacturer's standard commercial rating practice for the size and model crane furnished.

3.3.19 Reduced load capacities chart. When specified (see 6.2), the crane shall meet or exceed the following requirements for reduced lifting requirements for nuclear lifts:

a. Controls shall ensure the load is under positive operator control.
b. Crane shall have features that ensure prompt and automatic stop except for cable swing in case of operating mechanism failure.
c. Design of the crane shall permit completely synchronized operations.
d. Crane shall have features that will automatically neutralize controls in the absence of a manual force.
e. The reduced lift capacity and positioning limits and rates of the crane shall be identified on a separate load chart based on the following criteria:

1. The crane shall have a minimum safety factor of three for all load sustaining (basic structure) parts, based on the yield strength of those parts.
2. The load lines shall have a minimum safety factor of five based on their ultimate strength.
3. Blocks and rope falls, fiber rope and webbing shall have a minimum safety factor of five based on ultimate strength.
4. Accessory parts (such as hooks, rings, shackles, slings and wire rope) shall have a safety factor of five based on ultimate strength.

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f. All load hooks shall be fitted with throat opening safety devices.

g. The crane shall have stops or limit switches to stop the wire rope when the hooks reach their respective upper limits.

h. No reference to nuclear certification shall be made on the crane markings or in the technical data.

3.3.20 Stress requirements. Design of the completely assembled crane shall be such that the design strength margins will not be less than the minimum strength margins contained in SAE J1063. Tests, in accordance with SAE J1063 and SAE J765 shall have been performed on the crane being provided and test results shall be made available for Government inspection at the time of first article tests.

3.3.21 Interchangeability. All units of the same classification, furnished with similar options under a specific contract, shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories and spare parts.

3.3.22 Servicing and adjustment. The contractor shall service and adjust all cranes for operational use, including at least the following: Adjustment of lights; adjustment of the brake systems and mechanical components of the crane engine, transmission and drive train; servicing of the cooling system with a 50-50 solution, by volume, of ethylene-glycol-type antifreeze conforming to MIL-A-46153 and water; and cleaning, bleeding and filling of the hydraulic system. The crane shall be completely lubricated as specified in 3.3.9.

3.3.23 Stenciling. The gross weight of each crane, with equipment, shall be stenciled on each side of the crane in such a manner as to be readily discernible to dock and military personnel. The prescribed tire pressure shall be stenciled on the carrier fenders by block or stencil-type letters not more than 1 inch high.

3.4 Maintainability. The crane design, together with component and accessory location and installation, shall permit ready accessibility to all items requiring periodic maintenance service in the field. Maintenance will be accomplished with the use of conventional general-purpose tools associated with equipment of this nature. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage requirements and minimum disturbance to other elements of the crane.

3.5 Transportability. The crane shall be capable of being embarked and disembarked, without disassembly, from all roll-on/roll-off (RO/RO) ships, LST 1139 and LPD classes of ships. It shall also be capable, under its own power, of embarking/disembarking through the surf from either the LCU, LCM-8 Landing Craft and LCAC.

3.6 Safety. The crane shall comply with applicable OSHA regulations and ANSI standard B30.5. All rotating or reciprocating parts and all parts subject to high operational temperatures, that are of such nature or are so located as to be or become a hazard to the operating or attending personnel, shall be substantially guarded, or insulated, to the extent necessary to eliminate the hazard. The principal platform walking surfaces shall be of an anti-skid type. Ladders, steps and handholds shall be provided in such quantity and of such size on the sides of the crane cab or superstructure that

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entrance thereto and exit therefrom may be unhampered and nonhazardous. Engine cooling fans shall have heavy grille or ring type guards. Electrical equipment shall be effectively guarded and grounded to protect personnel from electrical shock hazard. The contractor shall apply the Government furnished electrical warning decals (see 3.11) to the vehicle in a prominent position.

3.7 Electromagnetic interference suppression. Electromagnetic radiation from the crane shall be within the limits of SAE J551. When required, the crane shall be equipped with radio noise suppressors conforming to SAE J552.

3.8 Dimensions and weight. The overall dimensions of the crane shall be:

Length Chassis	40 Feet (Maximum)
Length Overall	45 Feet (Maximum)
Width	102 Inches (Maximum)
Height	144 Inches (Maximum)
Boom Retracted	35 Feet (Maximum)
Boom Fully Extended	80 Feet (Minimum)

The axle loads of the crane shall be no more than:

Steering Axle(s)	20,000 Pounds Each
Rear Axle(s)	22,000 Pounds Each
Gross Crane Weight	75,000 Pounds (Maximum)

3.9 Treatment and painting. All interior and exterior portions of the crane that are normally painted and all exposed surfaces that require protection from the environment shall be cleaned, treated and painted in accordance with MIL-T-704, type F for ferrous metals and type G for nonferrous metals and fiberglass top coat, color green 383. Primer in accordance with MIL-P-53030 may be used in lieu of MIL-P-52192. Finish coat in accordance with MIL-C-53039 (polyurethane) may be used in lieu of MIL-C-46168. Vendor parts and components received by the prime contractor with either an alkyd primer or alkyd paint coating may be top coated with the above specified primer and paints applied directly over existing alkyd primer or paint without additional surface preparation except for cleaning if the base alkyd primer or paint meets applicable Military/Federal specifications and is allowed to cure for a period of 14 days. This exception for vendor parts and components is not intended to supersede the requirement that the complete crane be primed and painted in accordance with MIL-T-704 using the primers and paints specified above. If another color is specified (see 6.2), the above system shall be overcoated with MIL-C-46168 or MIL-C-53039 of the desired color. All surfaces of the crane which exceed 400°F during operation shall not be primed or painted as specified above, but shall be painted in the appropriate color of the base coat or camouflage pattern scheme if applicable, with paint conforming to either MIL-P-14105 for steel surfaces or TT-P-28 for aluminum surfaces.

3.10 Vehicle marking. Registration numbers and other markings shall be applied in accordance with MIL-STD-642 except that camouflage pattern painted cranes shall have the markings in black segments painted camouflage green 383 and the markings in green and brown segments painted black. Lifting and tiedown attachments shall be identified with 1-inch letters indicating "SLING POINT" or "TIEDOWN". The tire pressure shall be marked on the frame near the

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wheels or wheel hubs with block or stencil-type letters not more than 1 inch and not less than 3/4 inch high.

3.10.1 Identification plate. An identification plate shall be furnished by the contracting officer for each crane. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose and securely affix a plate to each crane in a conspicuous place with nonferrous screws, rivets or bolts not less than 1/8 inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.10.2 Instruction plates. The crane shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws, rivets or bolts of not less than 1/8 inch diameter.

3.11 Government-furnished property. The following property shall be furnished by the Government for installation on the crane by the contractor (see 6.4):

a. Electrical Warning Decal Set, NSN: 7690-00-267-5778 - one each
b. Vehicle Weight Classification sign set, NSN: 9905-00-565-6267 - one each
c. Vehicle Identification Plate - one each

3.12 Government-loaned property. The following property shall be loaned to the contractor for use in testing the crane (see 6.5):

a. STE/ICE, NSN: 4910-00-124-2554 for tests (see 4.6.14 and 4.6.15)
b. DCA - Tester

3.13 Workmanship

3.13.1 Steel fabrication. Steel used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.

3.13.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lock washers shall be provided in accordance with good commercial practice, and all bolts, nuts and screws shall be tight.

3.13.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivets. Rivet heads shall be full, neatly made, concentric with the rivet holes and in full contact with the surface of the member.

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3.13.4 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Welding shall conform to the procedures as prescribed by either the AWS or ASME for the type of welding operation to be performed.

3.13.5 Machine work. Tolerances and gages for metals fits shall conform to the limitations specified herein and on the applicable drawings, and otherwise to the standards of good commercial practice.

3.13.6 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required, the first article shall be subjected to the tests specified in 4.6.1 through 4.6.15. Failure to pass any test shall be cause for rejection. Upon acceptance of the first article, the first article shall remain at the manufacturing facility as a production sample and shall be the last crane delivered on the contract. The first article shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first article in a serviceable condition for the duration of the contract.

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4.4 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.5, the tests of 4.7 and the packaging inspection of 4.9.

4.5 Examination. Each crane shall be examined for compliance with the requirements specified in Section 3. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.6 Tests

4.6.1 Crane safe load capacity test. The crane shall be completely serviced and subjected to the manufacturer's recommended break-in. The safe load capacity test shall be performed on a level, paved surface by connecting a Baldwin SR4 load cell, or equal, between the lifting hook and a dead weight, or the direct use of calibrated weights. The crane shall then lift the loads specified in 3.3.14 and 3.3.18. The crane, with safe load raised, shall be rotated through the rated working range at a safe operational speed. Actual tipping conditions shall be determined in accordance with SAE J765. Eighty-five percent of each determined tipping load shall be compared with the applicable safe load. If the crane manufacturer has previously conducted this test on an identical model crane, the certified test records shall be made available to the Government inspector and the contracting officer during the first article testing and may be acceptable in lieu of conducting the test.

4.6.2 Stress test requirements. The crane shall be tested in accordance with SAE J1063 to ensure conformance with 3.3.20. The contractor may submit certified records of test performance for this test.

4.6.3 Speed tests. Speed tests shall be taken over a 1 percent grade on a paved road. Time shall be recorded with a stopwatch accurate to 1 second. The average speed shall be recorded for the unit. Maximum average speed not less than that specified in 3.3 when calculated by the following formula shall be attained:

$$V = \frac{0.682 \times D}{T}$$

Where: V = speed in mph
D = distance in feet
T = time in seconds

A fifth wheel type recorder may be used.

4.6.4 Gradeability. The crane shall be tested for gradeability in compliance with 3.3. The crane, without load and with mounted hook block, shall negotiate a 45 percent slope on paved road at not less than 1 MPH.

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4.6.5 Power load lowering test. General inspection of the powerdown system shall be made for adequate lubrication, ease of operation of controls, safety precautions and for good manufacturing practices. The power load lowering system shall be tested as follows to determine conformance to 3.3.16:

a. With the crane equipped with a boom of a length as specified in 3.3.1.2, and with outriggers set, position the revolving superstructure in the most stable craning position and assemble a single line load equal to the maximum rated load divided by the permissible line pull.
b. Position the boom angle for the minimum working radius.
c. Hoist the load 20 or more feet above the ground level, or until interference with the crane boom is obvious.
d. Prepare the crane for powerdown operation on the load line.
e. Operate the power load lowering system in accordance with manufacturer's instructions, and lower the load under power. Vary the speed of the load.
f. When the load reaches the ground, immediately repeat operations c., d, and e. These operations shall be repeated until 20 cycles have been completed.

4.6.6 Road test. The crane shall be given a road test for a distance of not less than 500 miles at speeds up to 45 MPH over paved roads; 14 miles over uneven roads at not less than 15 MPH; and 4 miles over unpaved or unprepared surfaces composed of sandy or loose soil, or both, at maximum safe speeds.

4.6.7 Travel control test. To demonstrate workability of the travel controls (see 3.3), the crane shall be operated forward and backward in two figure-8 patterns and hard right and left turns.

4.6.8 Service brake test. The service brakes of the crane shall be tested for compliance with 3.3.5.3. The service brakes shall be applied bringing the crane to a complete halt to determine conformance to 3.3.5.3. Immediately after the crane comes to a halt, this test cycle shall be repeated for four additional cycles. All shoes, drums and other brake components, where there is indication of damage, shall be disassembled and examined.

4.6.9 Parking brake test. The parking brake shall be tested for compliance with 3.3.5.3.

4.6.10 Crane hook test. The manufacturer is required to furnish a certification from the crane hook manufacturer certifying that the hook furnished with the crane is of the compatible size, having the proper load capacity and safety factor without defects or discontinuities. When this certification is not furnished, the crane hook shall be tested in the following manner. The crane hook shall be given a wet or dry ferro-magnetic particle test to determine surface or subsurface cracks or discontinuities. Either method of test used shall show clearly visible particles defining existing discontinuities. Any discontinuities shall be cause for rejection (see 3.3.1.4).

4.6.11 Lifting and tiedown attachments test. The lifting and tiedown attachments shall be tested for conformance to 3.3.13. The contractor may submit certified calculations and drawings in lieu of this test with the approval of the contracting officer.

4.6.12 Measurement of electromagnetic radiation. To determine conformance to 3.7, electromagnetic radiation shall be measured in accordance with SAE

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J551. The manufacturer may, upon approval of the contracting officer, furnish a certification in lieu of the test that the crane meets the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.6.13 Attachments. When attachments are specified (see 6.2), the crane shall be tested to show the capability to install and operate the attachment in the allotted time (see 3.3.10).

4.6.14 DCA capability. The contractor shall perform a 100 percent visual inspection of all DCA wiring harnesses prior to installation on the crane to insure wire and connections are free of error. After installation on the crane, test the DCA harness (reference: Chapter 5 of Design Guide for DCA) on each vehicle with a Government loaned DCA tester prior to interfacing with STE/ICE. Nonconformance to 3.3.6.5 shall constitute failure of the DCA.

4.6.15 DCA-STE/ICE interface. Upon completion of inspection and tests specified in 4.6.14, test the DCA for all measurements specified in 3.3.6.5 using the Government loaned STE/ICE. After completion of first article testing, this test shall be repeated and the results compared to the first examination performed as specified in 4.5. Nonconformance to 3.3.6.5 or failure of any component or wiring in the DCA harness which results in inability to obtain the measurements previously recorded with the STE/ICE (4.5) shall constitute failure of this test.

4.7 Production unit operational test. Each production crane shall be completely assembled, adjusted, lubricated, or otherwise serviced for operation. The engine shall be started and subjected to a warm-up period as recommended by the manufacturer. The crane shall be given a run-in test and all controls operated a sufficient number of times to ascertain that the components and mechanisms actuated by the controls operate promptly, fully, and without restrictions or malfunction. The manufacturer's standard production tests shall then be conducted on each production crane.

4.8 Test records. Contractor's records of all first article tests and inspections, giving the results of said tests and inspections to determine compliance with the requirements of this specification, shall be kept complete and available to the Government representative and the contracting officer. Test and inspection records shall be signed and approved by a person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory, satisfactory to the Government, capable of conducting required tests to determine compliance with all the requirements specified herein.

4.9 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of Section 4 of MIL-C-3580. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame specified (see 6.2).

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5. PACKAGING

5.1 Preservation and packaging. Preservation and packing shall be level A, B or C, as specified (see 6.2).

5.1.1 Level A or B. Preservation and packing shall be in accordance with MIL-C-3580, except that antifreeze conforming to MIL-A-46153 shall be used.

5.1.2 Level C. Preservation and packing shall provide protection against deterioration and damage and assure carrier acceptance and delivery to destination, without damage, at the lowest ratings of the carrier in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules. Gearboxes and crankcases shall be filled with operational lubricants and the cooling system shall be filled with a fresh clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153. Marking shall reflect as a minimum, the item name, national stock number, quantity, contract number and appropriate address markings. The markings shall be applied with a waterproof material, be legible and shall provide a definite contrast.

5.2 Marking. The cranes and attachments, interior packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cranes are intended for use in beach staging areas, unloading cargo from lighterage; clamshell, grapple, concrete bucket and pile driving operations; and general lifting on construction sites.

6.2 Ordering data

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. When first article is required (see 3.1).
c. When salt spray protection is required (see 3.3.2).
d. When tires shall be high flotation type (see 3.3.5.5).
e. When winterization kit shall be provided (see 3.3.5.8 and 3.3.7).
f. When additional STE/ICE-DCA test parameters are required (see 3.3.6.5).
g. When attachments are required (see 3.3.10).
h. Pile driving system specification (see 3.3.10.3).
i. When vehicle weight classification sign kit is required (see 3.3.11).
j. When reduced lift capability is required (see 3.3.19).
k. When cleaning treatment and painting shall be other than that specified (see 3.9).
l. When a preproduction pack inspection is required and time frame required for submission (see 4.9).
m. Level of preservation and level of packing required (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data

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Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID No.
4.3	First Article Inspection	DI-T-4901
4.3	First Article Inspection Report	DI-T-4902

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractor in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item to be tested should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 3.1. The first article should consist of one unit. The contracting orricer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.11.

6.5 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.12.

6.6 Subject term (key word) listing.

Boom
Bucket
Clamshell
Crane
Grapple
High-mobility
High-speed
Hydraulic
Multi-axle
Revolving

Preparing activity:
Navy - MC
Project No. 3810-N100

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-29425(MC)	CRANE: HIGH-SPEED, HIGH-MOBILITY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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